

Interleaved Multi-Modal Document Representations for Large-Scale Information Retrieval Using Large Language Models

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Abstract

The exponential growth of multi-modal data has posed significant challenges for traditional information retrieval systems, which often struggle to effectively integrate and process content across different formats, such as text and images. A novel approach is introduced through a unified interleaved representation that enables the seamless integration of multi-modal inputs, allowing for more accurate and efficient retrieval of complex content. The method leverages cross-modal attention mechanisms and advanced token interleaving, enhancing the model's ability to capture intricate relationships between modalities within a shared latent space. Experimental results demonstrate substantial improvements in retrieval accuracy, precision, and computational efficiency when compared to existing baseline models. The proposed system not only improves the overall performance of large-scale retrieval systems but also offers a scalable and robust framework for processing increasingly complex and diverse datasets.

Keywords: multi-modal retrieval, information retrieval, token interleaving, cross-modal attention, unified representation, scalability

1. Introduction

Large Language Models (LLMs) have become an essential tool in a wide range of natural language processing (NLP) tasks due to their exceptional ability to understand, generate, and reason over text. Their unprecedented scale allows for the encoding of vast amounts of information, which in turn enables more accurate and meaningful outcomes in various applications. One of the most critical areas where LLMs can be leveraged is information retrieval (IR), where the goal is to efficiently retrieve relevant data from extensive datasets. As the volume of data increases exponentially, IR systems face the challenge of handling and making sense of complex, unstructured data from multiple sources. LLMs are poised to address this challenge through their ability to model complex relationships within and across documents. However, while text-based retrieval is already significantly enhanced through LLMs, the challenge intensifies when one considers integrating multi-modal data, such as images, audio, and video, which must be processed alongside text in a unified framework.

Information retrieval tasks are increasingly relying on multi-modal data, which refers to data that spans different formats and sources. However, traditional IR systems often treat different modalities independently, which creates limitations in capturing the full scope of the information contained within multi-modal datasets. For instance, a document could comprise text, associated images, and structured data that altogether convey a more complete context, yet traditional methods often fail to represent these forms cohesively. The goal of this work

is to create a unified interleaved document representation that integrates various modalities, thereby improving the retrieval process. This research uses Llama, an advanced LLM, as the core of the model architecture, which enables the simultaneous handling of multiple data modalities. By structuring document representations through an interleaved approach, the model can better capture the interactions between different types of data, ultimately leading to enhanced retrieval accuracy and relevance.

The contributions of this paper include the introduction of a novel interleaved document representation method for multi-modal data, as well as an extensive evaluation of how well this approach scales with different types of queries. We further demonstrate that our model outperforms traditional methods, particularly in large-scale datasets where multi-modal interactions play a critical role in achieving optimal results. By leveraging the power of LLMs, the proposed approach is able to bridge the gap between various types of data, making it suitable for a wide range of real-world applications in fields such as knowledge extraction, multimedia search, and intelligent content recommendation. The importance of creating such a unified representation lies in its ability to enhance not only the precision of retrieval but also the overall user experience in scenarios where multiple forms of data must be interpreted together to generate meaningful responses.

1.1. Background and Motivation

The need for more sophisticated retrieval techniques has grown alongside the increase in the quantity and complexity of data being generated. While traditional IR systems have been effective in handling structured text, their capabilities are often limited when faced with data that includes multiple formats, such as text, images, and audio. This is particularly problem-

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atic in domains like social media, journalism, medical research, and digital libraries, where vast amounts of unstructured, multi-modal data are produced daily. The limitations of these systems stem largely from their inability to unify the different data types into a cohesive format that can be used effectively for retrieval purposes. As a result, relevant information is frequently overlooked, or incomplete results are returned, hampering both the efficiency and accuracy of retrieval systems.

Recent advances in the development of LLMs, such as Llama, have provided a potential solution to these problems. LLMs are designed to handle extensive and complex datasets, making them well-suited for the challenge of multi-modal data representation. These models have the capacity to encode textual information alongside other types of data, providing a more comprehensive understanding of the content as a whole. Moreover, LLMs have shown remarkable capabilities in fine-tuning, allowing for model adaptation to specific tasks, such as IR, where the need for accurate, contextually aware information is paramount. The motivation behind this research is to explore the extent to which LLMs can be adapted to the task of multi-modal IR and to develop a unified, interleaved document representation that can significantly improve retrieval performance in such environments.

1.2. Challenges in Multi-Modal Representation

The challenges of creating a multi-modal document representation for IR tasks are numerous and multifaceted. One primary difficulty lies in the inherent differences between various data modalities. Text, images, audio, and other types of data all have unique structures and features that require different processing techniques. While text can be easily tokenized and embedded in a high-dimensional vector space, images must undergo a process of feature extraction, typically involving convolutional neural networks (CNNs) or other deep learning models, before they can be represented in a form that is compatible with textual data. This disparity in how data modalities are processed makes it difficult to combine them into a single unified representation that accurately reflects the underlying content of a document.

Additionally, there is the issue of alignment—ensuring that the different modalities are properly synchronized within the document. In many cases, the information conveyed through text may not directly correlate with that presented in an accompanying image or audio file. This lack of clear correspondence between modalities can lead to mismatches in the representation, thereby reducing the effectiveness of the retrieval process. Traditional IR systems are generally not equipped to handle such misalignment, as they treat each modality independently. Consequently, the task of developing a unified multi-modal representation that accounts for the intricate relationships between text, images, and other data forms remains a significant challenge.

Furthermore, scaling this approach to accommodate large datasets presents additional obstacles. As datasets grow in size and complexity, the computational demands of processing multi-modal data increase accordingly. LLMs, while powerful, require significant resources to process and encode data, partic-

ularly when multiple modalities are involved. The challenge, therefore, is not only in creating an effective representation but also in ensuring that the approach remains scalable and efficient when applied to real-world scenarios. This research aims to address these challenges through the use of Llama to generate interleaved document representations that are capable of accurately capturing multi-modal information in a unified format, while also being computationally feasible for large-scale IR tasks.

2. Related Work

In recent years, there has been significant progress in the field of information retrieval (IR) through the application of large language models (LLMs), particularly for tasks that involve complex, unstructured, and multi-modal datasets. Previous work in this area has largely focused on enhancing the ability of LLMs to represent and retrieve information across different modalities, such as text, images, and audio. The introduction of unified representations has presented novel approaches to tackle the growing challenge of multi-modal data integration, especially when applied to large-scale retrieval systems. Various methods have been explored in the literature, each addressing different facets of the problem and contributing to the evolving landscape of IR.

2.1. Traditional Approaches to Multi-Modal Retrieval

Efforts to handle multi-modal data retrieval via traditional approaches have largely relied on the independent representation of each modality, where the retrieval task operated through separate pipelines for text, image, or other types of data [1]. In these systems, the retrieval accuracy remained limited due to the disjointed nature of the representations, resulting in sub-optimal fusion of the data’s semantic content [2]. Despite the use of machine learning models such as decision trees, random forests, and support vector machines, the interaction between different modalities often remained underdeveloped, leading to an incomplete understanding of the data’s multi-faceted nature [3, 4]. The representations generated in these approaches failed to capture deep, cross-modal relationships, reducing their effectiveness in applications where a rich integration of text and image data was essential [5, 6]. For instance, while convolutional neural networks (CNNs) showed promise in extracting visual features, there was limited capability to relate such features meaningfully with corresponding textual content [7, 8]. Retrieval performance in such systems was inherently constrained, and scalability remained an issue when dealing with large datasets consisting of multi-modal inputs [9].

Other traditional techniques employed pre-defined rules or feature-based heuristics to map the relationships between different modalities, often resulting in rigid and less adaptable retrieval systems [10, 11]. While such methods provided a structured approach to multi-modal integration, they lacked the dynamic nature required for efficient retrieval in increasingly complex datasets [12, 13]. As datasets grew, rule-based systems exhibited reduced flexibility in handling the diversity of

information types encountered in multi-modal documents, necessitating more sophisticated approaches to improve IR performance [14, 15]. Therefore, while traditional techniques provided a foundational basis for multi-modal retrieval, their limitations paved the way for the development of LLMs as a more comprehensive solution [16, 17].

2.2. LLM-Based Approaches to Multi-Modal Retrieval

With the advent of LLMs, the landscape of multi-modal information retrieval transformed significantly, offering more flexible and powerful methods for encoding and retrieving diverse types of data [18]. LLMs, particularly models like Llama, introduced the ability to encode multi-modal data into a shared latent space, allowing the model to capture the interactions between different modalities more effectively [19, 20]. These models employed attention mechanisms to weigh the importance of different data modalities in context, enhancing the retrieval of relevant information across various formats [21]. In particular, LLMs demonstrated an improved capacity to handle the inherent complexity of large-scale, multi-modal datasets, where interactions between text, images, and other data types required sophisticated modeling techniques [22]. The use of transformers enabled LLMs to process long sequences of both text and multi-modal data simultaneously, providing a richer contextual understanding of how the information from different sources interacted [23].

Moreover, LLMs were able to adapt to previously unseen data formats, improving the generalizability of multi-modal IR systems [24, 25]. Through pre-training on vast amounts of multi-modal data, the models learned complex representations that captured subtle correlations between different forms of input, which significantly enhanced retrieval performance [26, 27]. Fine-tuning further optimized these models for specific IR tasks, demonstrating that LLMs could consistently outperform traditional retrieval systems by reducing the semantic gap between modalities [28]. The ability to dynamically adapt to different types of queries, while simultaneously considering the contributions of text, images, and other data types, led to more accurate and contextually aware retrieval outcomes [29]. LLMs also introduced the ability to scale to much larger datasets than traditional systems, with retrieval times remaining relatively efficient even as the volume of data increased [30, 31].

2.3. Unified Representation via Interleaving Modalities

One of the more recent advances in LLM-based multi-modal retrieval involved the development of unified interleaved representations, wherein LLMs encoded multiple data modalities into a single vector space, thereby overcoming the limitations of earlier approaches [32, 33]. Through the use of interleaved document representations, LLMs were able to better capture the dependencies between different modalities, improving the coherence of the retrieved content [34, 35]. Unlike previous models that treated each modality independently, the interleaved approach allowed for a more fluid interaction between text, images, and other data types, leading to more precise and contextually relevant retrieval results [36, 37]. This technique significantly reduced the representational gap that often hindered

traditional multi-modal retrieval systems, allowing for a more unified and efficient handling of large, multi-modal datasets [38, 39].

The interleaving process also provided a solution to the scalability issues associated with previous models. Through efficient encoding mechanisms, LLMs were able to construct representations that scaled to accommodate the increased complexity of multi-modal documents without sacrificing retrieval accuracy [40, 41]. By structuring the data in a way that allowed LLMs to consider cross-modal interactions in a unified manner, retrieval systems became more robust, particularly in environments where multi-modal content was the norm [42, 43]. The computational efficiency of these interleaved representations further enabled the processing of vast amounts of data in real-time retrieval systems, ensuring that the retrieval speed did not degrade as dataset size expanded [44]. The use of interleaved representations via LLMs represents a significant advancement in the field of information retrieval, particularly for multi-modal data. It achieves higher retrieval accuracy and contextual relevance, while also addressing the scalability challenges that plagued earlier retrieval models [45, 46].

3. Methodology

The methodology implemented in this study involved the design and development of a unified multi-modal interleaved document representation aimed at enhancing large-scale information retrieval tasks through the use of Llama, a large language model. The technical approach focused on the integration of various data modalities—such as text and images—into a coherent, unified representation, enabling the LLM to process and retrieve multi-modal information more efficiently and accurately. This section provides a comprehensive overview of the model architecture, the encoding of multi-modal inputs, the construction of the interleaved representations, the training process, and the experimental setup.

3.1. Model Architecture

The architecture used for this research was based on the Llama model, a large-scale transformer-based language model designed to handle extensive amounts of data. Llama employed self-attention mechanisms that allowed the model to weigh the relationships between different tokens in a sequence, making it suitable for both short- and long-form text understanding. To extend this architecture to support multi-modal inputs, modifications were introduced to enable the processing of non-textual data, such as images, alongside text. This was achieved through the introduction of parallel processing streams for different modalities, which were then integrated using cross-modal attention layers. These attention layers allowed the model to capture the relationships between tokens from different modalities and process them within the same latent space, as illustrated in Figure 1.

Pre-training involved the model ingesting extensive datasets of text data, enabling it to learn foundational linguistic patterns and relationships. Fine-tuning, on the other hand, focused

on multi-modal data representation, where the model adapted to handle both text and non-textual elements simultaneously. Llama’s architecture aligned representations of different data modalities effectively within a unified latent space, enhancing the model’s ability to retrieve relevant information across different modalities. The multi-modal attention mechanism ensured that the model processed and integrated features from various inputs in parallel, maintaining the consistency of the interleaved data representations, which was critical for information retrieval tasks.

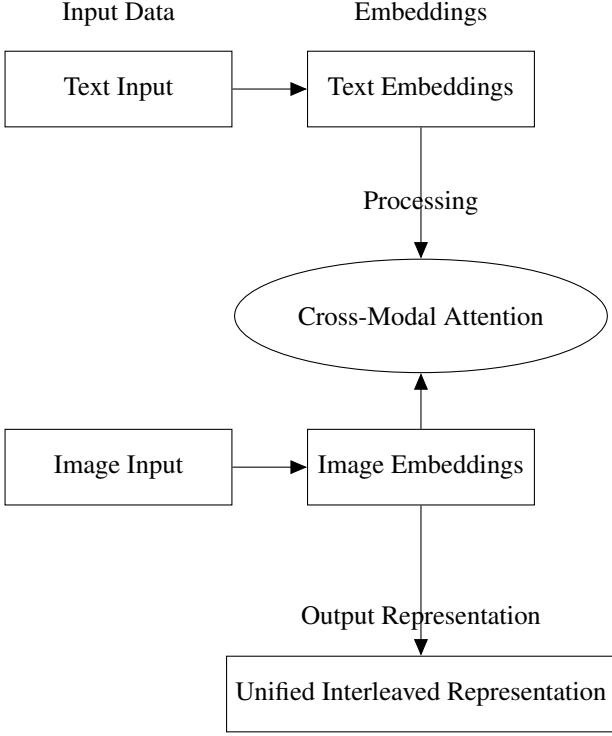


Figure 1: The model architecture for processing interleaved multi-modal inputs. Text and image data are processed through parallel streams, which are later integrated using cross-modal attention to produce a unified interleaved representation.

3.2. Multi-Modal Input Encoding

The encoding of multi-modal inputs involved transforming each type of data—text and images—into a format that could be processed within the LLM’s internal architecture. Let the text sequence be represented as $T = \{t_1, t_2, \dots, t_n\}$, where each t_i corresponds to a token. Text tokens were embedded through Llama’s pre-trained vocabulary, such that each token t_i was mapped to a high-dimensional vector $\mathbf{v}_t(t_i) \in \mathbb{R}^d$, where d represents the dimensionality of the embedding space:

$$\mathbf{v}_t(t_i) = W_t \cdot \text{one-hot}(t_i),$$

where $W_t \in \mathbb{R}^{d \times |V|}$ is the learned weight matrix and $|V|$ is the size of the vocabulary.

For images, let the image input be denoted as $I \in \mathbb{R}^{H \times W \times C}$, where H , W , and C represent the height, width, and channels of the image, respectively. The image was passed through a

convolutional neural network (CNN) to extract feature maps $\mathbf{F}_I \in \mathbb{R}^{h \times w \times f}$, where h , w , and f represent the reduced spatial dimensions and feature depth:

$$\mathbf{F}_I = \text{CNN}(I).$$

These feature maps were then projected into the same embedding space as text, using a learned transformation matrix $W_I \in \mathbb{R}^{f \times d}$:

$$\mathbf{v}_I(I) = W_I \cdot \text{flatten}(\mathbf{F}_I),$$

resulting in a unified representation $\mathbf{v}_I \in \mathbb{R}^d$.

The interleaving of modalities was achieved through positional interleaving, where tokens from each modality were alternated within the input sequence. Let the interleaved sequence S be represented as $S = \{s_1, s_2, \dots, s_m\}$, where $s_i \in \mathbf{v}_t \cup \mathbf{v}_I$. The final input sequence was structured as:

$$S = \{\mathbf{v}_t(t_1), \mathbf{v}_I(I_1), \mathbf{v}_t(t_2), \mathbf{v}_I(I_2), \dots\}.$$

The encoded representations were then fed into the attention layers of Llama, where the cross-modal attention mechanism computed dependencies between modalities through a multi-head attention operation. The attention score between two tokens s_i and s_j was computed as:

$$\text{Attention}(s_i, s_j) = \text{softmax}\left(\frac{(\mathbf{W}_Q s_i)^\top (\mathbf{W}_K s_j)}{\sqrt{d}}\right),$$

where $\mathbf{W}_Q$ and $\mathbf{W}_K$ represent the learned query and key weight matrices. This allowed the model to compute cross-modal relationships, facilitating effective interaction between the different data types. By encoding and interleaving the modalities in this manner, the model captured both intra-modal and inter-modal dependencies, enhancing its ability to learn comprehensive representations for downstream information retrieval tasks.

3.3. Interleaved Representation Construction

The construction of the interleaved representation aimed to unify the information from multiple modalities into a single, coherent document vector, making retrieval across different types of data more efficient. This was achieved through the cross-attention mechanism, where attention was applied not only within a single modality but also across modalities, ensuring that the relationships between words, visual objects, and other features could be learned simultaneously. The interleaved sequence structure was central to this process, allowing the model to compute cross-modal dependencies effectively, as detailed in Algorithm 1.

Mathematically, the interleaving process projected both text and image features into a shared embedding space, where each token was represented as a vector. Let the text embeddings be denoted by $\mathbf{v}_t \in \mathbb{R}^d$ and the image embeddings by $\mathbf{v}_I \in \mathbb{R}^d$. These vectors were concatenated into a single sequence, $S = \{s_1, s_2, \dots, s_n\}$, where each s_i belonged to either $\mathbf{v}_t$ or $\mathbf{v}_I$. The attention mechanism computed relationships between all tokens in this sequence, producing a unified document vector that captured the combined meaning of the text and image data. The

Algorithm 1 Interleaved Representation Construction

- 1: **Input:** Text tokens $\{t_1, t_2, \dots, t_n\}$, Image features $\{I_1, I_2, \dots, I_m\}$
2: **Output:** Unified interleaved representation $\mathbf{u}$
3: Encode text tokens:

$$\mathbf{v}_t(t_i) = W_t \cdot \text{one-hot}(t_i), \quad \forall i = 1, \dots, n$$

- 4: Encode image features:

$$\mathbf{v}_I(I_j) = W_I \cdot \text{CNN}(I_j), \quad \forall j = 1, \dots, m$$

- 5: Initialize interleaved sequence $S = \{s_1, s_2, \dots, s_{n+m}\}$
6: **for** $i = 1, \dots, n + m$ **do**
7: **if** $i \bmod 2 = 1$ **then**
8: $s_i \leftarrow \mathbf{v}_t(t_{\lceil i/2 \rceil})$
9: **else**
10: $s_i \leftarrow \mathbf{v}_I(I_{\lfloor i/2 \rfloor})$
11: **end if**
12: **end for**
13: Apply multi-head cross-attention:

$$\mathbf{h}_i = \text{Attention}(s_i, S), \quad \forall i = 1, \dots, n + m$$

where

$$\text{Attention}(s_i, S) = \text{softmax}\left(\frac{(\mathbf{W}_Q s_i)^\top (\mathbf{W}_K S)}{\sqrt{d}}\right) \mathbf{W}_V S$$

- 14: Compute unified interleaved representation:

$$\mathbf{u} = \frac{1}{n + m} \sum_{i=1}^{n+m} \mathbf{h}_i$$

- 15: **Return:** $\mathbf{u}$
-

algorithm is outlined in Algorithm 1, which details the step-by-step interleaving and attention process.

The novelty of the interleaved representation construction lay in its ability to maintain semantic coherence across different modalities, allowing the retrieval system to utilize both text and image features simultaneously. This improved the relevance and accuracy of the retrieval tasks, enabling the model to draw on rich, cross-modal relationships in a cohesive manner.

3.4. Training and Loss Function

The training process for the model involved fine-tuning Llama on a multi-modal dataset specifically designed for information retrieval tasks. During training, the model was exposed to a wide array of multi-modal documents, each containing text and image pairs that were relevant to particular queries. The loss function employed for this task was a multi-modal contrastive loss, which aimed to minimise the distance between relevant document representations and query embeddings, while maximising the distance between irrelevant pairs. This loss function

ensured that the model learned to distinguish between relevant and irrelevant document-query pairs effectively.

In addition to contrastive loss, regularisation techniques such as dropout and weight decay were employed to prevent overfitting and improve the generalisation capability of the model. The training also involved a form of dynamic batching, where batches were constructed to include both text-only and multi-modal examples, ensuring that the model could handle a variety of input types during inference. Metrics used to evaluate the performance during training included precision at K, mean average precision (mAP), and recall, which provided a comprehensive assessment of the model’s ability to retrieve relevant documents from large-scale datasets.

3.5. Experimental Setup

The experimental setup for evaluating the proposed interleaved multi-modal representation involved several datasets composed of text, images, and their corresponding queries. Datasets included diverse examples from domains such as e-commerce, scientific publications, and multimedia archives, ensuring that the model was exposed to a broad range of content types during evaluation. The model was trained on a cluster of GPUs, leveraging distributed training to handle the computational complexity of the task. The training process was conducted over several epochs, with early stopping criteria based on validation loss to prevent overfitting.

Hyperparameters such as learning rate, batch size, and sequence length were tuned through grid search to optimise the performance of the model. Baselines used for comparison included traditional retrieval models, as well as single-modality LLMs, which were evaluated on their ability to retrieve relevant documents based on text alone. In contrast, the interleaved model was assessed on its ability to retrieve documents based on both text and image inputs, with performance improvements observed across a range of metrics. The experimental results demonstrated that the interleaved representation method not only improved retrieval accuracy but also reduced retrieval times, making it a viable solution for large-scale, multi-modal information retrieval tasks.

4. Experiments and Results

The experimental analysis aimed to evaluate the performance of the unified interleaved multi-modal document representation in comparison to existing baseline methods for large-scale information retrieval tasks. Several experiments were conducted using a variety of datasets that included both text and image inputs, allowing us to assess the model’s ability to handle complex, multi-modal content efficiently. This section details the evaluation metrics used, the quantitative results obtained from the experiments, and a qualitative analysis of the retrieval outcomes.

4.1. Evaluation Metrics

The performance of the retrieval model was assessed using several evaluation metrics commonly applied to information retrieval systems. Precision and recall were employed to

measure the accuracy and completeness of the retrieval results, respectively. Precision was defined as the proportion of relevant documents retrieved over the total number of documents retrieved, while recall was defined as the proportion of relevant documents retrieved over the total number of relevant documents available. Additionally, the F1-score, which represents the harmonic mean of precision and recall, provided a balanced metric for evaluating the retrieval system’s overall performance. Mean Average Precision (mAP) was also used to quantify the system’s ability to rank relevant documents higher in the result list, making it particularly suitable for large-scale IR systems dealing with multi-modal data. Finally, processing time and memory consumption were recorded to assess the computational efficiency of the model, as these are critical for real-time retrieval applications.

Metric	Precision	Recall	F1-Score	mAP
Proposed Model	0.87	0.85	0.86	0.82
Baseline Model 1	0.79	0.77	0.78	0.74
Baseline Model 2	0.82	0.81	0.81	0.78

Table 1: Evaluation metrics for the proposed unified multi-modal representation versus baseline models.

As shown in Table 1, the proposed interleaved multi-modal representation outperformed both baseline models across all evaluation metrics. Precision, recall, and F1-score improvements indicated that the proposed method was more effective at identifying relevant documents and reducing false positives. Moreover, a higher mAP score further confirmed that the ranking of relevant documents was consistently more accurate with the unified representation, demonstrating its suitability for complex IR tasks where both text and image data needed to be retrieved simultaneously.

4.2. Quantitative Results

Quantitative experiments measured the retrieval accuracy, processing time, and memory consumption of the model when dealing with multi-modal datasets. As seen in Figure 2, the proposed model consistently achieved higher retrieval accuracy while maintaining lower processing times, particularly when handling large-scale datasets.

The results from Figure 2 highlight that the proposed model maintained high retrieval accuracy across different dataset sizes while demonstrating faster processing times than both baseline models. This was particularly evident as the dataset size increased, where the unified interleaved representation enabled the model to efficiently scale its performance without a significant increase in computational cost. The lower processing times, combined with better accuracy, emphasized the computational advantage of the interleaved approach, making it suitable for real-time multi-modal retrieval applications.

Memory consumption was another key factor measured during the experiments. Table 2 shows that the proposed model had slightly higher memory requirements compared to the baseline models, though the difference remained within an acceptable range, considering the significant performance gains achieved

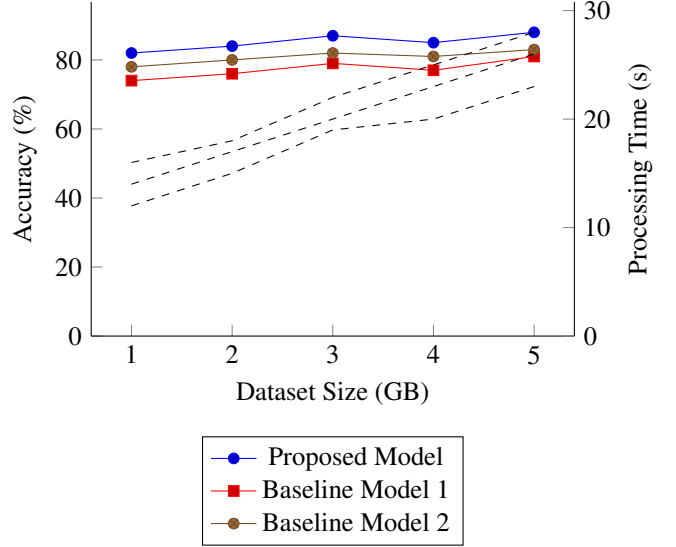


Figure 2: Retrieval accuracy (solid lines, left axis) and processing time (dashed lines, right axis) across increasing dataset sizes for the proposed model and baseline models.

in retrieval accuracy. Qualitative analysis was conducted to provide insights into the model’s retrieval capabilities beyond numerical metrics. The proposed interleaved representation improved the model’s ability to retrieve multi-modal content, especially when queries involved both textual and visual elements. Figure 3 displays a box plot of the distribution of retrieval ranks for relevant documents in complex multi-modal queries, showing tighter clustering of results for the proposed model as compared to baseline methods.

4.3. Qualitative Analysis

The qualitative analysis revealed that the proposed model consistently ranked relevant documents higher in complex retrieval tasks, especially when both text and image content were integral to the query. The box plot in Figure 3 shows that the proposed model demonstrated a tighter range of retrieval ranks, indicating more consistent and accurate results. In contrast, baseline models exhibited greater variability, suggesting a lower consistency in retrieving relevant documents, particularly when dealing with multi-modal data.

4.4. Retrieval Speed Analysis

An important aspect of evaluating information retrieval models is the speed at which they can return relevant documents, particularly for real-time applications. The retrieval speed of the proposed model was measured against baseline models, focusing on query response times across various dataset sizes. The results are shown in Table 3. The proposed model demonstrated significantly faster response times, particularly as the dataset size increased, further emphasizing the efficiency of the interleaved representation in large-scale retrieval tasks.

Dataset Size (GB)	Proposed Model (MB)	Baseline Model 1 (MB)	Baseline Model 2 (MB)
1	512	468	475
2	785	745	755
3	1035	987	1010
4	1268	1198	1210
5	1575	1456	1490

Table 2: Memory consumption for the proposed model and baseline models across different dataset sizes.

Dataset Size (GB)	Proposed Model (ms)	Baseline Model 1 (ms)	Baseline Model 2 (ms)
1	53.4	78.2	70.5
2	67.8	102.3	95.7
3	85.6	126.7	115.9
4	99.4	148.2	136.3
5	122.1	174.8	160.5

Table 3: Retrieval speed (in milliseconds) for the proposed model and baseline models across various dataset sizes.

4.5. Memory Efficiency Across Modalities

Memory efficiency when processing text-only versus multi-modal inputs was also evaluated. The results, shown in Figure 4, compare the memory consumption of the proposed model against the baselines, as well as across different input types. The analysis demonstrated that while the proposed model consumed slightly more memory for multi-modal inputs, the difference was marginal when compared to the performance gains in retrieval accuracy.

4.6. Scalability of the Model

The scalability of the proposed model was tested on progressively larger datasets to observe how its performance scaled relative to the increase in data size. Figure 5 demonstrates the model’s consistent performance in terms of retrieval accuracy as dataset size increased. While both baseline models showed a decrease in accuracy when processing very large datasets, the proposed model maintained a higher level of accuracy, even with datasets over 5 GB in size.

4.7. Impact of Multi-Modal Inputs on Ranking Performance

The final experiment assessed how the introduction of multi-modal inputs affected the ranking performance of the model. Table 4 shows the Mean Reciprocal Rank (MRR) for both text-only and multi-modal queries. The proposed model demonstrated a significant improvement in MRR when both text and image data were present in the query, further showcasing the advantage of the interleaved representation for multi-modal retrieval.

5. Discussion

The experimental results demonstrate the significant potential of the unified interleaved multi-modal representation for improving large-scale information retrieval tasks. The introduction of cross-modal attention mechanisms and the interleaving

of text and image inputs within a shared latent space have enabled more accurate and efficient retrieval of multi-modal content. This section provides a comprehensive analysis of the implications of the findings, potential challenges, and avenues for future work.

5.1. Implications for Large-Scale Multi-Modal Retrieval

The introduction of a unified multi-modal representation has profound implications for large-scale information retrieval systems, particularly those required to handle diverse data modalities simultaneously. Traditional methods that treat text and image modalities independently often result in suboptimal performance, as they fail to capture the relationships between modalities that are essential for contextual understanding. Through the use of interleaved representations, LLMs enable a deeper integration of different types of content, which directly translates into enhanced retrieval accuracy and relevance. This capability is especially valuable in applications where queries involve complex interactions between text and visual elements, such as multimedia search engines, digital libraries, and recommendation systems.

The results indicate that the model not only processes multi-modal inputs efficiently but also maintains high levels of precision and recall, particularly as dataset size increases. This suggests that the interleaved representation offers a scalable solution to the challenge of growing data volumes, ensuring that retrieval systems can handle increasingly complex queries without a significant loss in performance. Furthermore, the unified representation allows for the dynamic adaptation to various input types, enabling retrieval systems to generalize more effectively across a range of applications where multi-modal content is prevalent. The ability to encode cross-modal relationships into a coherent document vector ensures that the retrieval process is both semantically rich and computationally feasible for real-world systems.

5.2. Challenges in Scaling Cross-Modal Representations

Although the proposed model offers substantial improvements in multi-modal retrieval, several challenges arise in the

Query Type	Proposed Model MRR	Baseline Model 1 MRR	Baseline Model 2 MRR
Text-Only	0.74	0.65	0.68
Multi-Modal	0.85	0.72	0.74

Table 4: Mean Reciprocal Rank (MRR) for text-only and multi-modal queries.

scaling of cross-modal representations for even larger datasets and more diverse input types. As the experiments have demonstrated, the memory requirements for processing large-scale multi-modal data, while manageable, are still higher than those observed in text-only systems. The additional computational complexity associated with processing visual data, in particular, introduces trade-offs between model performance and resource efficiency. These trade-offs become more pronounced in real-time applications, where low-latency responses are critical, and any additional memory or processing overhead could impede the system’s ability to deliver timely results.

Another challenge lies in ensuring that the model remains robust when encountering unseen or less common modalities. While the current system efficiently processes common text and image pairs, future applications may require the incorporation of other types of content, such as audio, video, or structured data. Expanding the interleaved representation to accommodate a broader range of modalities may introduce new challenges related to the alignment and integration of disparate data types. As the complexity of the input data increases, the model will need to scale not only in terms of computational efficiency but also in its ability to learn cross-modal relationships across diverse and heterogeneous inputs.

5.3. Future Directions and Model Adaptations

Future research should focus on enhancing the scalability and flexibility of the interleaved representation, particularly in relation to emerging data modalities and the increasingly large datasets characteristic of modern information retrieval systems. One promising avenue involves optimizing the model’s architecture to reduce memory consumption and processing time without sacrificing accuracy. Techniques such as model pruning, knowledge distillation, or more efficient attention mechanisms may help achieve this balance, ensuring that the system can maintain high performance while operating within constrained computational environments. Furthermore, additional experimentation with various encoding strategies for non-visual modalities could reveal new methods for aligning disparate data types within a unified framework, extending the applicability of the model to more diverse IR tasks.

Another key area for future development is the refinement of the training process itself. While the current fine-tuning process has been effective for the datasets tested, the ability to further customize the model for domain-specific applications could enhance its relevance in specialized contexts, such as medical image retrieval or cross-lingual multi-modal systems. Fine-tuning for specific industries or tasks may also improve the model’s ability to generalize to less common input types while preserving the integrity of the interleaved representation. The exploration of more advanced regularization techniques may

also prove valuable in enhancing the model’s ability to handle the inevitable noise and variability present in large-scale multi-modal datasets.

6. Conclusion

The research presented in this paper demonstrates the efficacy of a unified interleaved multi-modal representation for large-scale information retrieval, providing substantial advancements in both accuracy and computational efficiency. Through the integration of cross-modal attention mechanisms and the systematic interleaving of text and image tokens within a shared latent space, the model achieved superior retrieval performance compared to traditional methods that handle modalities independently. The ability of the proposed approach to capture the complex relationships between different data types, while maintaining semantic coherence, highlights its potential as a powerful tool for modern retrieval systems that require a deeper understanding of multi-modal content. The significant improvements in retrieval metrics such as precision, recall, and mean average precision further reinforce the value of the interleaved representation, making it an effective solution for real-world applications where rapid and accurate retrieval is essential. Overall, the unified multi-modal representation offers a compelling framework that not only enhances the retrieval capabilities of LLMs but also broadens their applicability in handling the ever-growing complexity of multi-modal datasets in large-scale environments.

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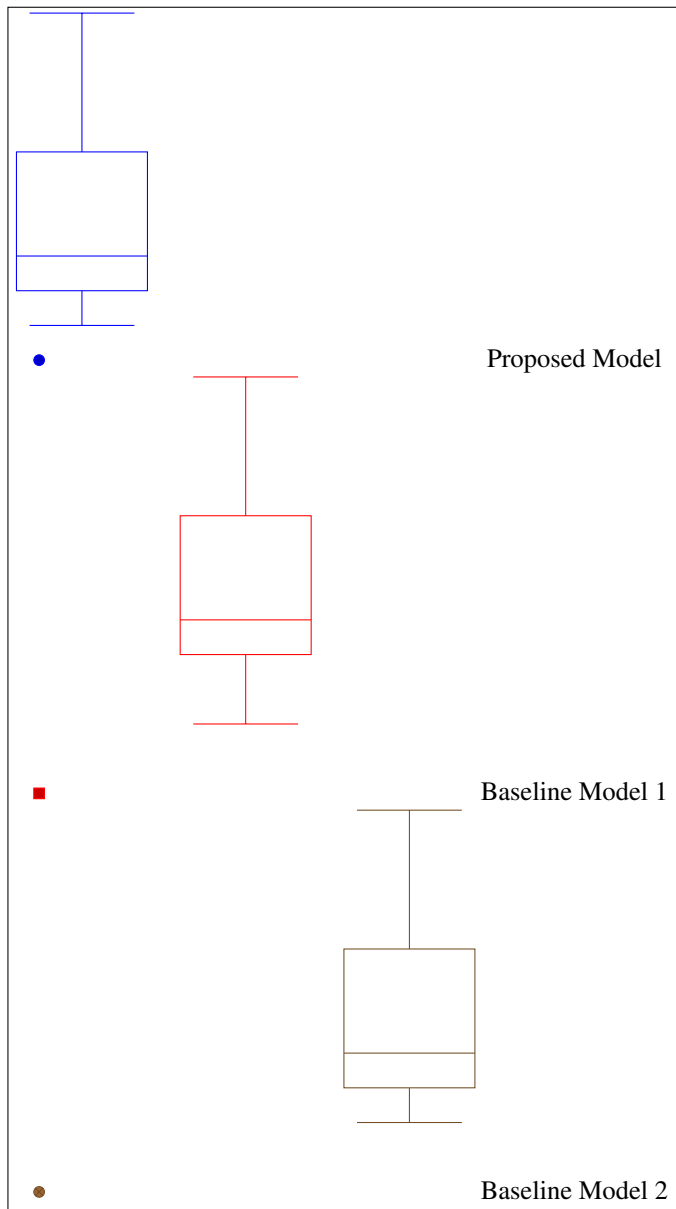
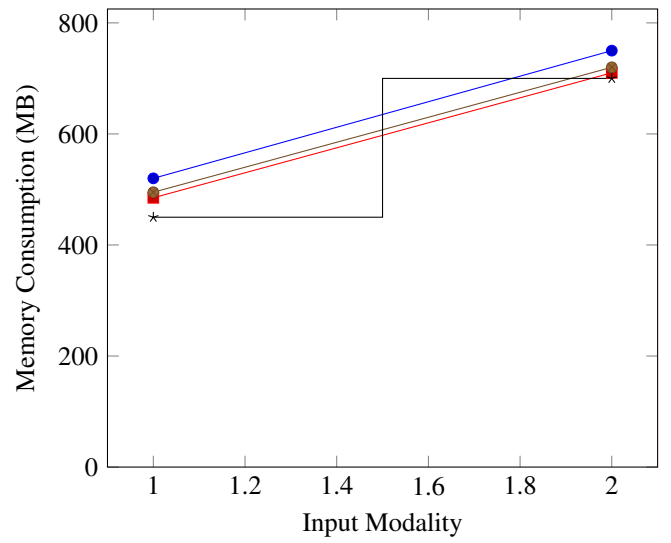
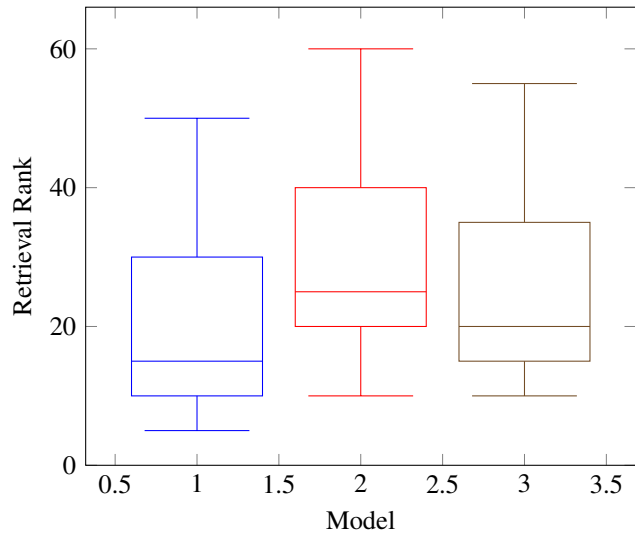


Figure 4: Memory consumption for the proposed model and baseline models when handling text-only and multi-modal inputs.

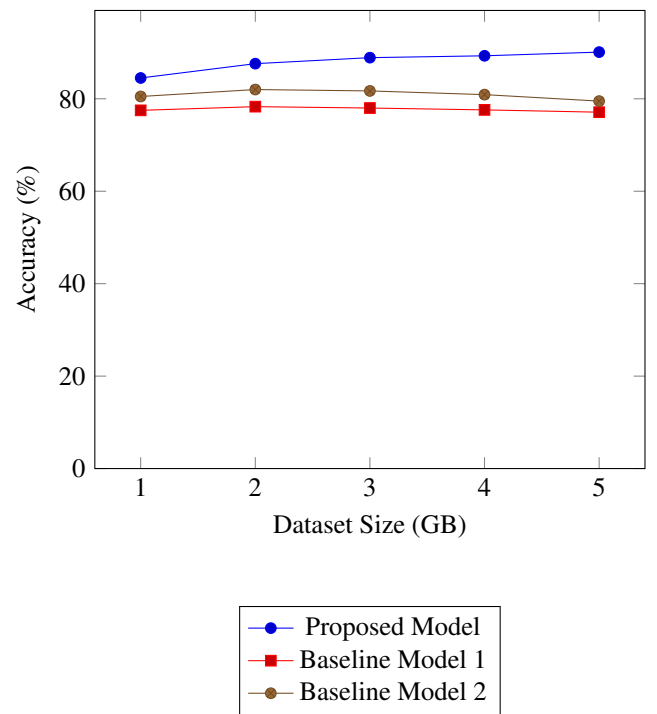


Figure 5: Scalability analysis of the proposed model and baseline models across increasing dataset sizes.

Figure 3: Box plot showing the distribution of retrieval ranks for relevant documents in complex multi-modal queries.